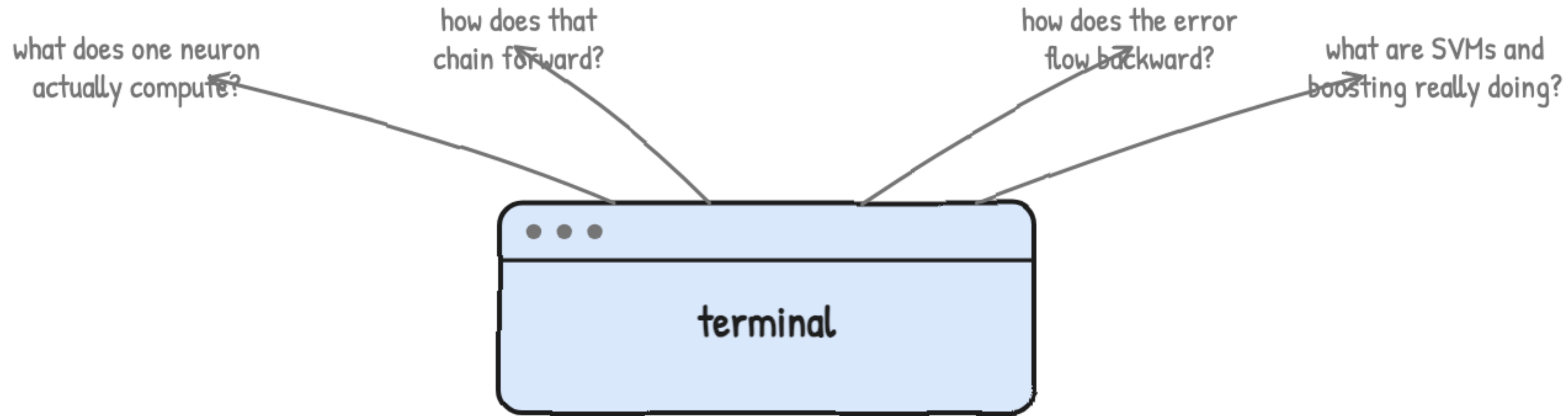


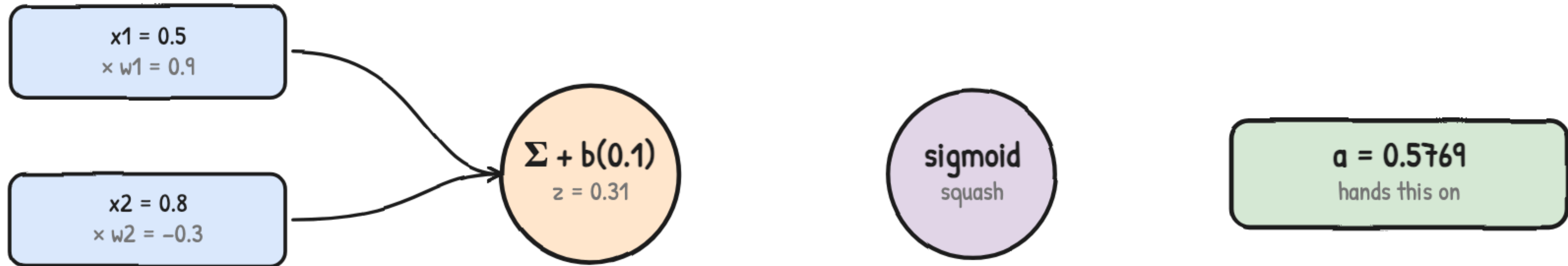
Four questions this deep dive answers



What a neuron computes, chains, and learns from

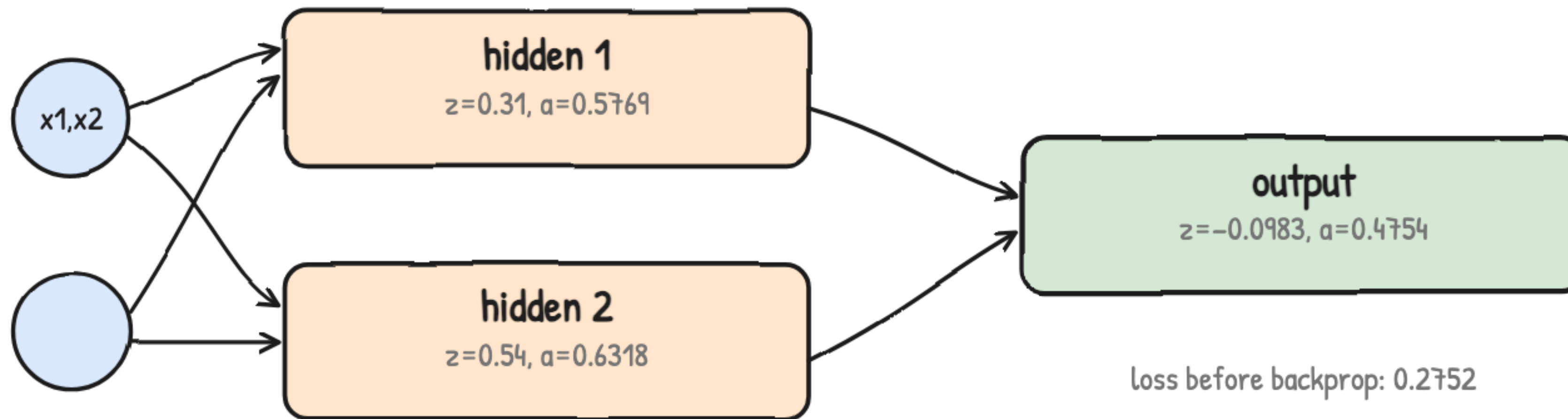
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One neuron does exactly two things to its inputs



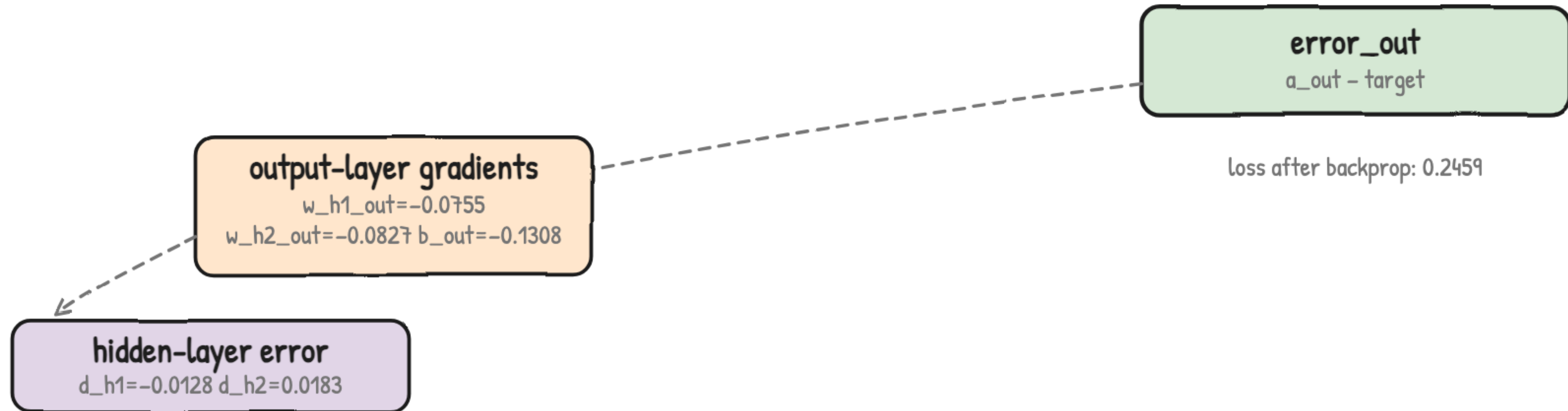
Weighted sum, then activation — on real numbers

One neuron's output becomes the next neuron's input



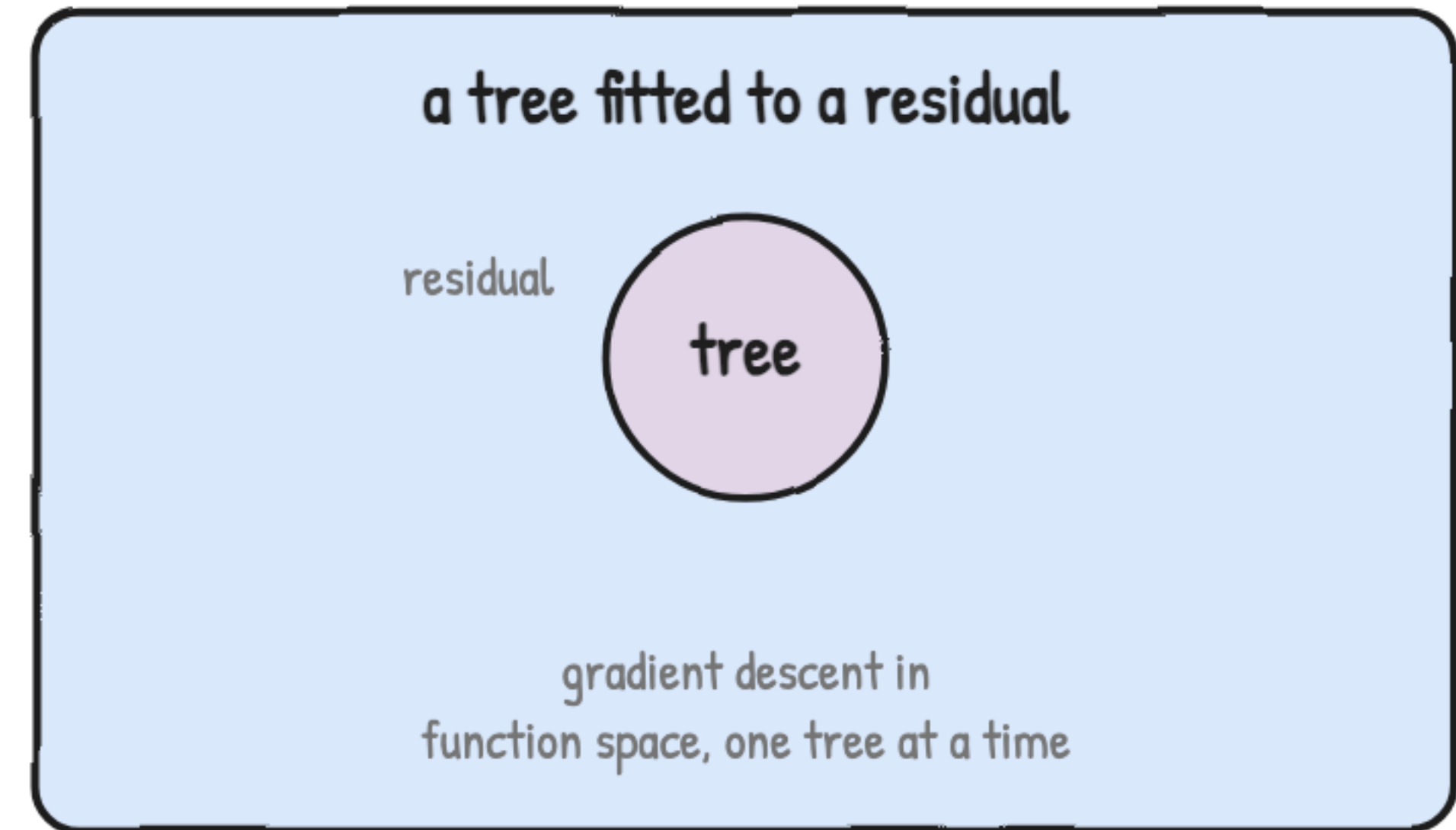
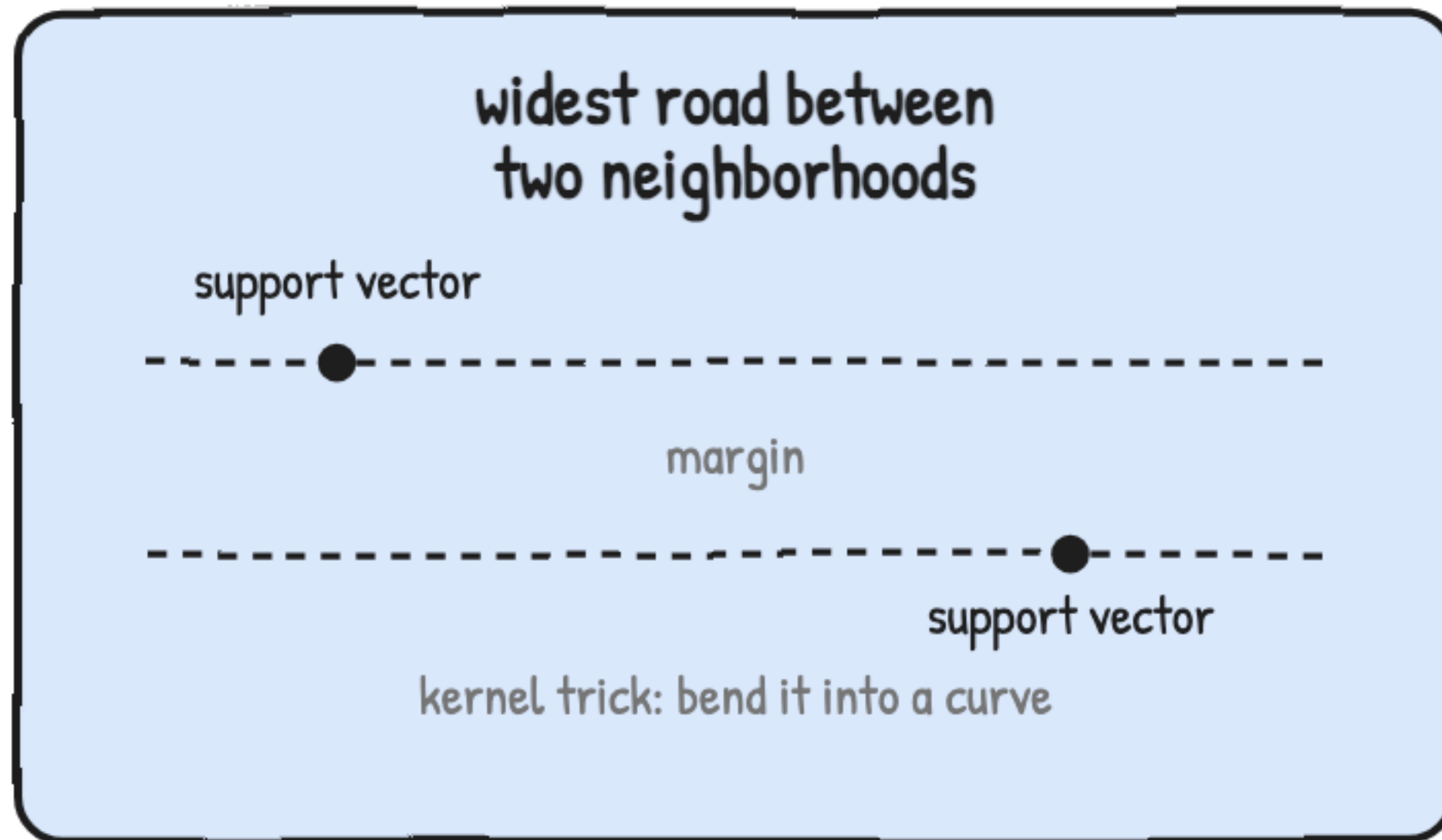
Hidden neuron 1's numbers are slide 2's numbers, exactly

The error flows backward, proportional to what actually moved it



A near-flat neuron barely moves the error backward through it

Two more toolbox extras: SVM and what boosting is really doing



A margin that doesn't move, and a tree fit to what's left over

Now: run the script, watch these exact numbers print

- 1 forward pass — hidden and output activations, loss
- 2 one backprop step — gradients, error signals, loss after

```
labs/m4 — python3 neuron_math.py
```

Same three-step sequence `model.fit()` ran for you · Gourav Shah · School of DevOps & AI